MCP Server Development

Class five

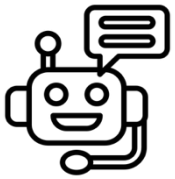
Agenda

- Overview of Agentic workflow
- Tools calling
- MCP
- Class exercise

Overview of AI Agent

What is AI Agent

From "answering questions" to "getting things done for you"



Traditional AI Solutions

- Answers your questions
- Waits for your instructions
- One step at a time
- You do most of the work







AI Agent

Brain (LLM)
Memory
Tools
Loop (iteration)

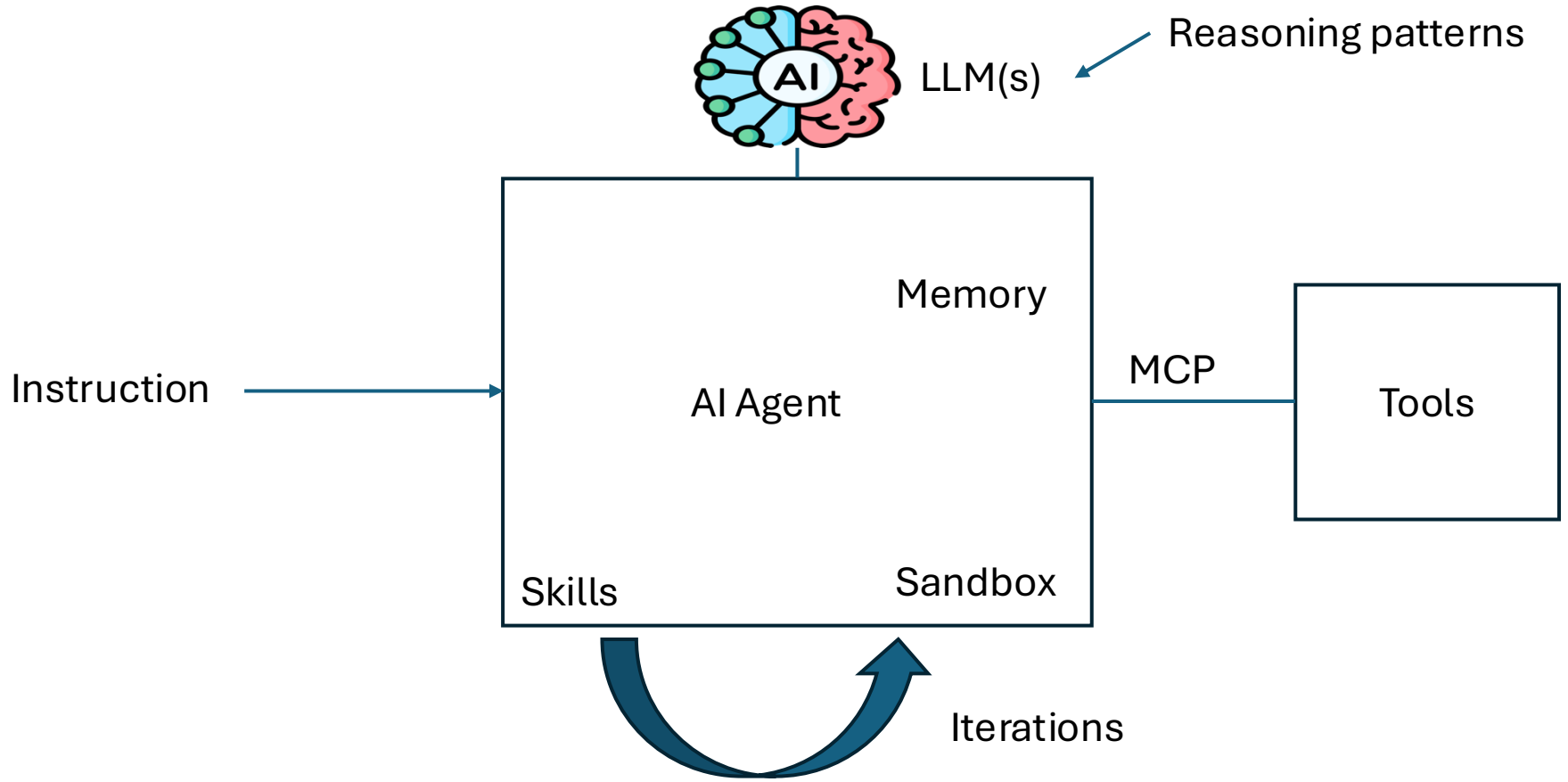
- Understands your goal
- Plans the steps itself
- Uses tools (email, calendar, web, apps)
- Takes action and finishes the task
- Learns and adapts as it goes

 Saves you time  Reduces your effort  Gets better results  Frees you up

Important Consideration (Risk)

- AI misunderstand the goal and make mistake
- Protect your privacy-Be careful what data you want to share
- Deploy Guardrail with 7x24 monitoring

Agentic Development Framework



AI Agent=LLM Model + Harness

How Agents Work — Two Core Patterns (LLM Reasoning Patterns)

ReAct

Reasoning + Acting

Think → Act → Observe → Repeat

- Interleaves thinking and action
- Highly adaptive to new results
- Best for exploratory / dynamic tasks
- Can get expensive on long chains

Plan-and-Execute

Plan first, then execute

Create full plan → Run steps → Replan if needed

- Planner creates a complete roadmap
- Executor carries out each step
- Better for long, structured tasks
- More predictable cost & structure

What is a Harness?

Harness = the engineering layer that makes the Agent reliable

Analogy

LLM = the horse (raw power & intelligence)
Harness = the bridle & gear (control, direction, safety)

Format Cleaning

Force clean structured output (e.g. valid JSON)

Parameter Validation

Check tool arguments before execution

Input / Output Filtering

Guard against injection & bad content

Code-level Corrections

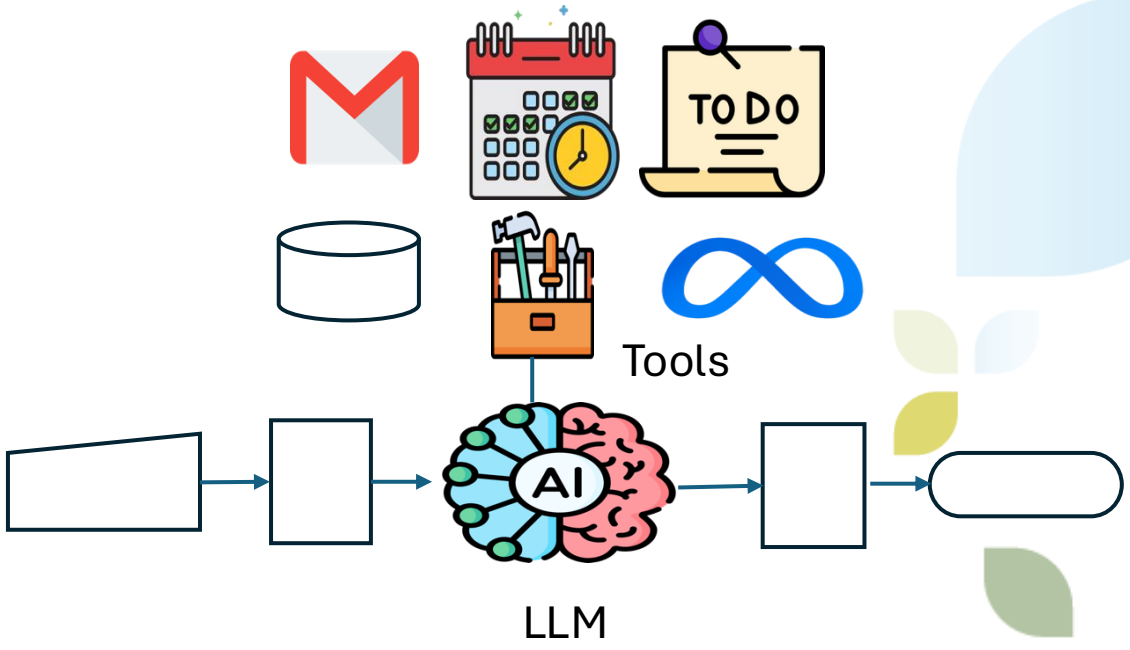
Hard rules that fix recurring mistakes

Agentic Workflow

Agentic Assistant

An AI System which could Plan and Act **Autonomously**

[User Goals]
↓
[Planner)] → Break down the task steps
↓
[Reasoner)] → Choose the appropriate action
↓
[Executor)] → Run the command or API
↓
[Memory] → [Knowledge Base)]
↓
[Feedback Loop] → Evaluate the results and improve



Example of Agentic Assistant



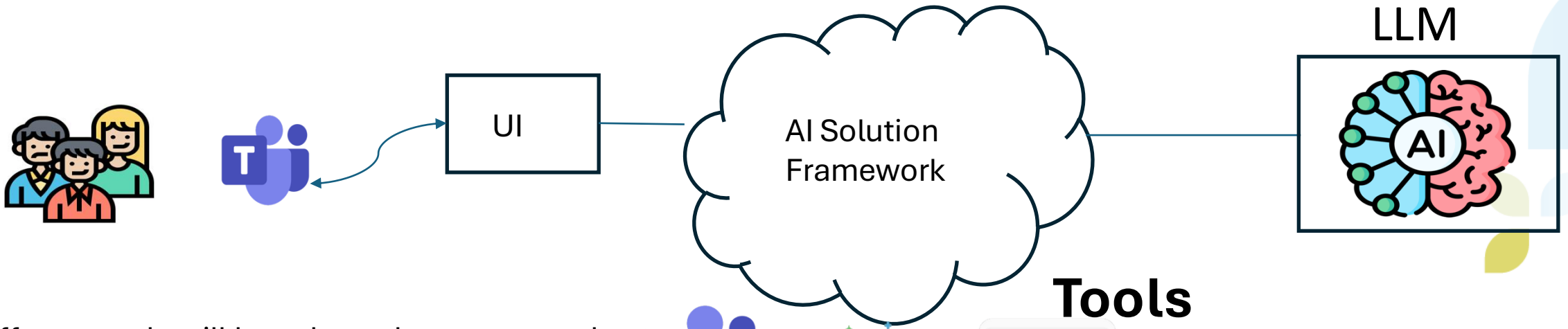
Send email to David and ask for the project update. Also send the meeting invite to him on next Friday 15:00

→

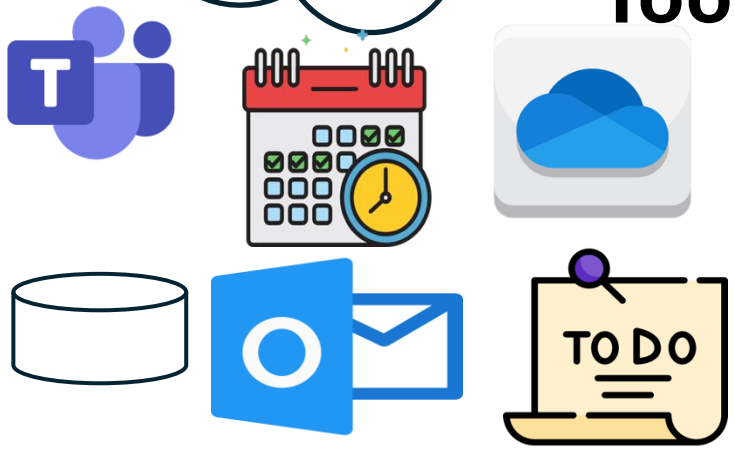
Task 1: Send an email to David requesting a project update.
Task 2: Send a meeting invite to David for next Friday at 15:00.

→

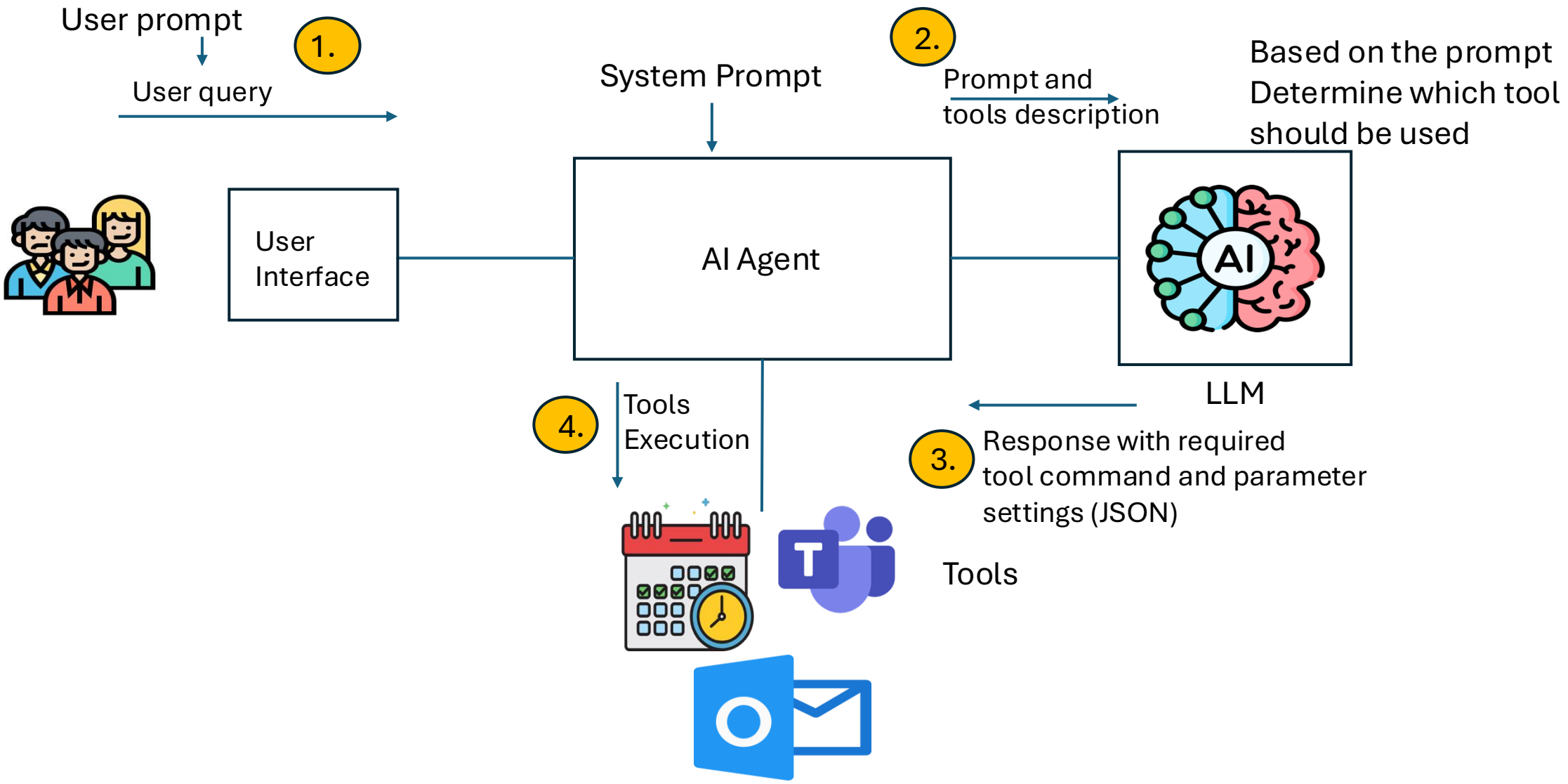
Select the required tools and plan/execute the tasks accordingly



Different tools will be selected autonomously based on the objectives of different sub-tasks



Tools Calling

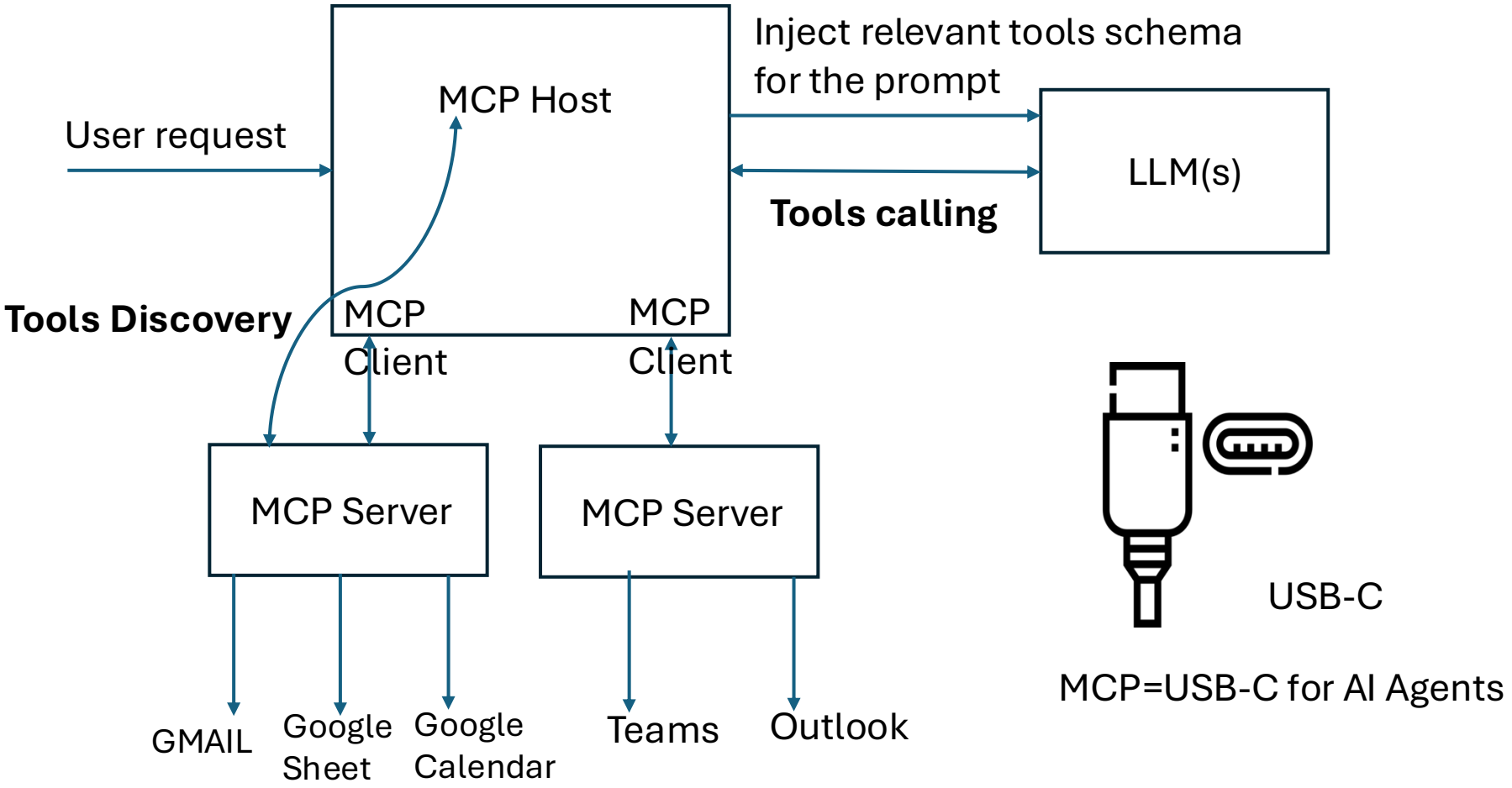


Model Context Protocol (MCP)

Components in MCP

Component	Role	Examples
MCP Host	The user-facing AI application (e.g., Claude Desktop, an IDE like Zed or Cursor, Copilot Studio, a custom agent app)	Chat interface, coding assistant
MCP Client	Embedded in the host; speaks the MCP protocol to servers	Handles requests from the LLM
MCP Server	Lightweight service that exposes tools, resources, or prompts	GitHub MCP server, SQL database server, local file system server

MCP (Model Context Protocol)
Standardize the way AI applications work with LLM



Benefits of MCP

MCP is the “USB-C for AI agents”:

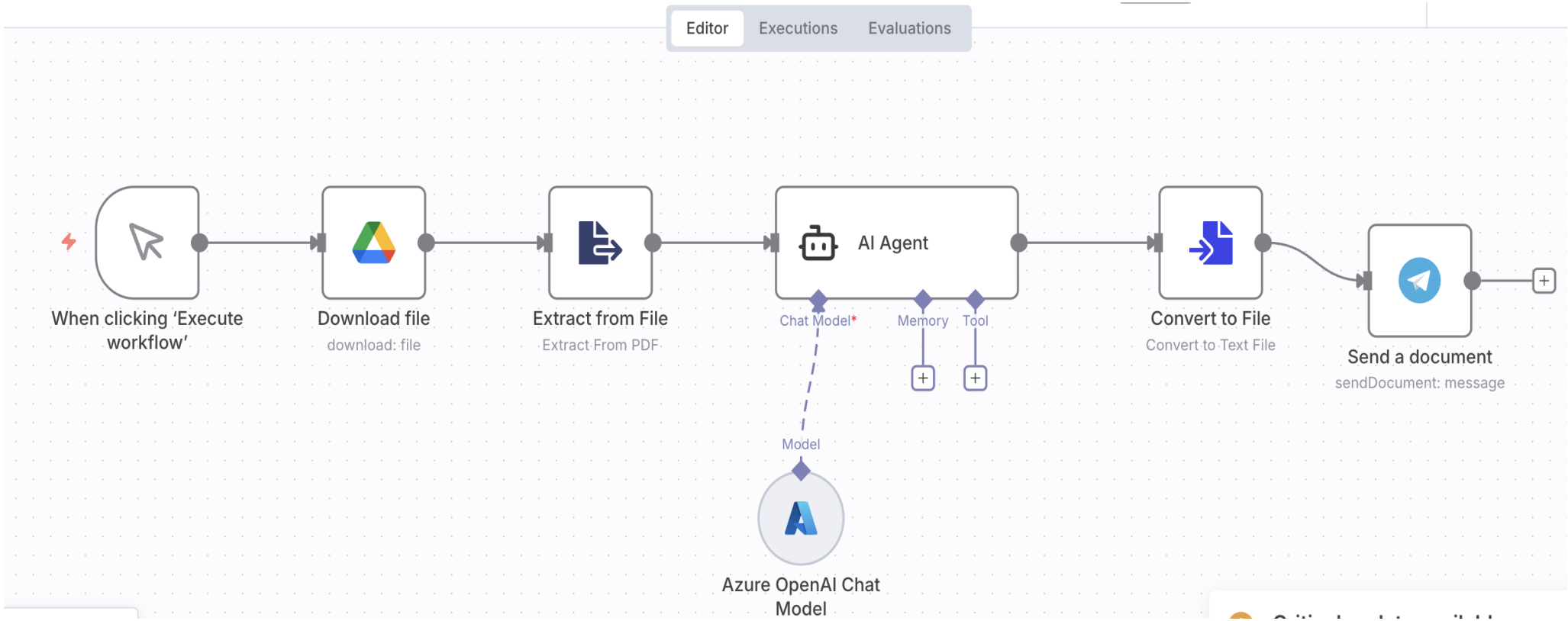
- Instead of building a custom cable for every new data source or LLM, you plug in standardized MCP servers, and everything just works — securely, privately, and across the entire AI ecosystem.
- One protocol to rule them all:
  - Instead of writing separate code for GitHub, Slack, Google Drive, a database, etc., you implement MCP once on the server side.
  - AI agents can dynamically discover, inspect, and use tools/resources without custom glue code.

Benefits of MCP

Benefit	Explanation	When It Matters Most
Modularity & Reusability	Tools live on separate servers. You can update, improve, or fix a tool without touching your agent.	Teams, production systems, long-term projects
Dynamic Tool Plugging	**Add new tools/servers on the fly without redeploying your workflow.**	Complex agents that need evolving capabilities
Standardization	One standard works across Claude, OpenAI, Grok, local models, n8n, etc.	Multi-model setups
Better Ecosystem	**Huge and growing library of ready-made MCP servers** (Git, Files, Browser, Databases, APIs, etc.)	Rapid development
Security & Control	**Tools run on their own servers with proper auth, permissions, and auditing.**	Enterprise / sensitive data
Discoverability	**Tools can describe themselves (capabilities, resources, prompts).**	Advanced agents
Separation of Concerns	**AI logic stays separate from tool logic. Different teams can own different parts.**	Large projects

Tools setup-n8n

n8n Workflow (Example)



n8n-Node Categories

Triggers

- Where, when and how of the agentic workflow begins

Actions

- Perform tasks in an app (e.g. upload file in Google drive)

Utility

- Transform data

Code

- Run code, send HTTP requests and work on webhooks etc

Advanced AI

- Langchain and LLM integration

Installation of n8n (local installation)

Please follow the installation procedure (attached word document)

Types of nodes in n8n-1/2

Category	Purpose	Examples of Nodes
Trigger Nodes	Start a workflow when something happens	Webhook Trigger, Schedule Trigger (Cron), Gmail Trigger (new email), Telegram Trigger (new message)
App / Action Nodes	Perform operations with specific apps & services	Google Sheets (add row), Gmail (send email), Slack (post message), Notion (create page), Airtable (update record)
Data Transformation	Clean, reshape, or restructure data	Set, Move/Remove Fields, Rename Keys, Item Lists, Split In Batches
Logic / Control Flow	Control workflow execution path	IF, Switch, Merge, Wait, Error Trigger
Function / Code	Add custom programming logic	Function (JavaScript), Function Item (per item JS), Code (JavaScript / Python)
AI & LangChain	AI reasoning & tool calling	OpenAI, Azure OpenAI, AI Agent (LangChain), Think Node, Embeddings
Utility Nodes	General-purpose helpers	HTTP Request, Webhook, Date & Time, Crypto, Spreadsheet File, File Operations
Database Nodes	Read/write structured data	MySQL, PostgreSQL, MongoDB, SQLite, Notion DB, Airtable
Messaging & Notifications	Communicate with users	Email (IMAP/SMTP/Gmail/Outlook), Slack, Teams, Telegram, Discord, Twilio (SMS/WhatsApp)

Types of nodes in n8n-2/2

Node	
Read/Write files from Disk	Import files(e.g. pdf/csv) to/from binary
Extract from file	Convert binary file to JSON
Convert to file	Convert JSON data to binary file
Edit Fields	Create/edit content (JSON)
AI Agent	AI framework
Memory	Memory module used in AI framework
LLM	e.g. Azure and other LLM
Web Hook	To receive http requests from other server
Command Node	To execute command (e.g. execute bat file under Windows OS)
HTTP Request	To send HTTP request to other nodes/server/AI Agent
Wait	To pend and wait for further message (e.g. web hook) or condition (time expired)
IF/SWITCH	Condition branching
Application nodes	Google/Meta/Telegram/Whatsapp/etc

Local n8n login account

User name: aiagent@fevaworks.com

Password: Feva3106

AI Agent with MCP Server

Use case: n8n+Python (Web service)